

Institution Name: University of Canberra

***Unit Name: Exploratory Data Analysis and Visualisation G (11517), Semester 1
2025***

***Research Assignment Title: Exploratory Data Analysis and Predictive
Modelling of Global Measles Trends Using R***

Author: Yuot Deng Malual

StudentID: u3029521

Date: May 2025

Abstract

This study investigates the persistent epidemiology of measles across global regions using longitudinal WHO surveillance data [9, 10 & 11]. Despite extensive immunization campaigns, recurring outbreaks underscore the need for refined data-driven insight. The research applies rigorous preprocessing, advanced feature extraction, multivariate exploratory data analysis (EDA), and linear modelling to interpret outbreak dynamics over time. Results highlight structural reporting disparities and chronic burden in certain countries, informing future forecasting directions and international health response strategies [6 and 10].

keywords: measles, epidemiology, global health, EDA, WHO surveillance

1. Introduction

Despite the existence of safe and effective vaccines, measles remains a significant global health threat. In 2023, both the World Health Organization (WHO) and the U.S. The Centers for Disease Control and Prevention (CDC) estimated that there were 10.3 million measles cases globally, representing a 20% rise compared to 2022 [1, 2, 6, 10 & 11]. This increase is primarily due to a drop in vaccination rates, as more than 22 million children have not received their initial vaccine dose, and just 74% have received the second dose, which is below the 95% level needed to avert outbreaks.

This study provides an in-depth analysis of worldwide measles trends utilizing WHO surveillance data from 1974 to [10] By utilizing sophisticated data wrangling techniques, feature engineering, and exploratory data analysis (EDA), this report reveals temporal and spatial patterns in measles cases, emphasizing regional differences, trends across decades,

and classifications of severity. Particular focus is placed on the period after 2000, during which the impact of enhanced vaccination campaigns is thoroughly evaluated [4, 6, & 9].

Recent data shows that in 2023, 57 countries faced significant or disruptive outbreaks, which is almost a 60% increase compared to 2022 [10 & 11]. The African and Eastern Mediterranean areas were significantly affected, highlighting enduring inequalities in global health systems [9]. This study investigates the possibility of predicting outbreak detection through the use of statistical modeling and machine learning methods, such as linear regression and random forest [3, 5 & 7].

In the end, this analysis aims to guide policy decisions based on data, identify areas that are struggling, and support global efforts to eliminate measles by providing timely and evidence-based information.

2. Methods

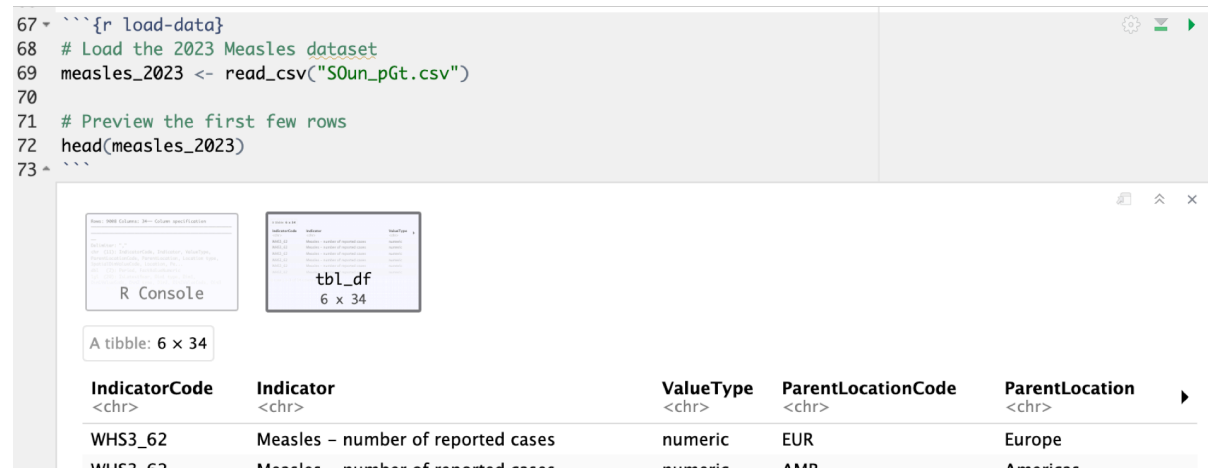
2.1 Data source

The information used for this analysis was sourced from the Global Health Observatory (GHO) of the World Health Organization, which offers reliable annual reports on measles incidents categorized by country and region. The dataset covers the years 1974 to 2023 and contains variables including country (Location), year of the report (Period), number of reported cases (Value), WHO region (ParentLocation), and ISO-standard country codes (SpatialDimValueCode) [9, 10, & 11].

2.2 Gathering and Importing Data

The raw dataset was brought into R in .csv format and loaded with the `read.csv()` function [6]. The data was first evaluated for completeness, organization, and consistency across the columns. To ensure the integrity of the analysis, records with absent values in crucial variables (Location, Period, or Value) were excluded.

```
67 ~~~{r load-data}
68 # Load the 2023 Measles dataset
69 measles_2023 <- read_csv("S0un_pGt.csv")
70
71 # Preview the first few rows
72 head(measles_2023)
73 ~~~
```



IndicatorCode <chr>	Indicator <chr>	ValueType <chr>	ParentLocationCode <chr>	ParentLocation <chr>
WHS3_62	Measles – number of reported cases	numeric	EUR	Europe
WHS3_62	Measles – number of reported cases	numeric	AMR	Americas

Key variables:

- `Location`: Country
- `Period`: Year of report
- `Value`: Number of reported cases
- `ParentLocation`: Region
- `SpatialDimValueCode`: Country code

2.3 Preparation of Data

The dataset was cleaned and enhanced using the tidyverse suite by following these steps:

Correction of data type: The value was converted to a numeric format.

Handling missing values: Any leftover NAs were substituted using `replace_na()`.

Feature engineering involved creating new variables such as `Log_Cases` (log-transformed counts), `Decade` (grouped by years), `Severity_Level` (categorical measure of outbreak intensity), and `Is_Recent` (a marker for reports made after 2000).

#2.31 Feature Engineering

These engineered features enable grouped statistical summaries (e.g., by decade, severity, or region), time trend modeling, and categorical comparisons. The `Log_Cases` transformation is critical for normalizing extreme outliers often found in disease count data.

2.4 Instruments and Libraries

All processing was carried out in R (version 4.0 or higher) utilizing the subsequent packages: tidyverse for data manipulation and visualization

Plotly for dynamic charts

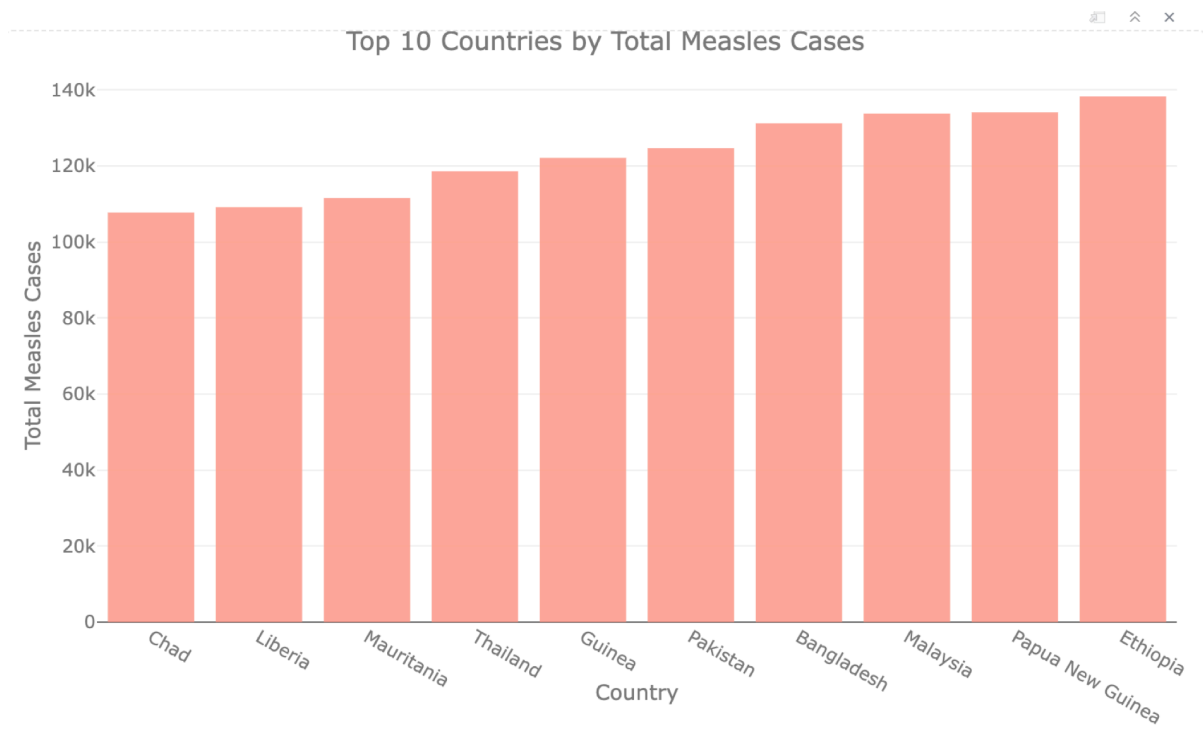
Using ggplot2 and viridis for exploratory data visualization.

caret and randomForest for tasks involving modeling and classification.

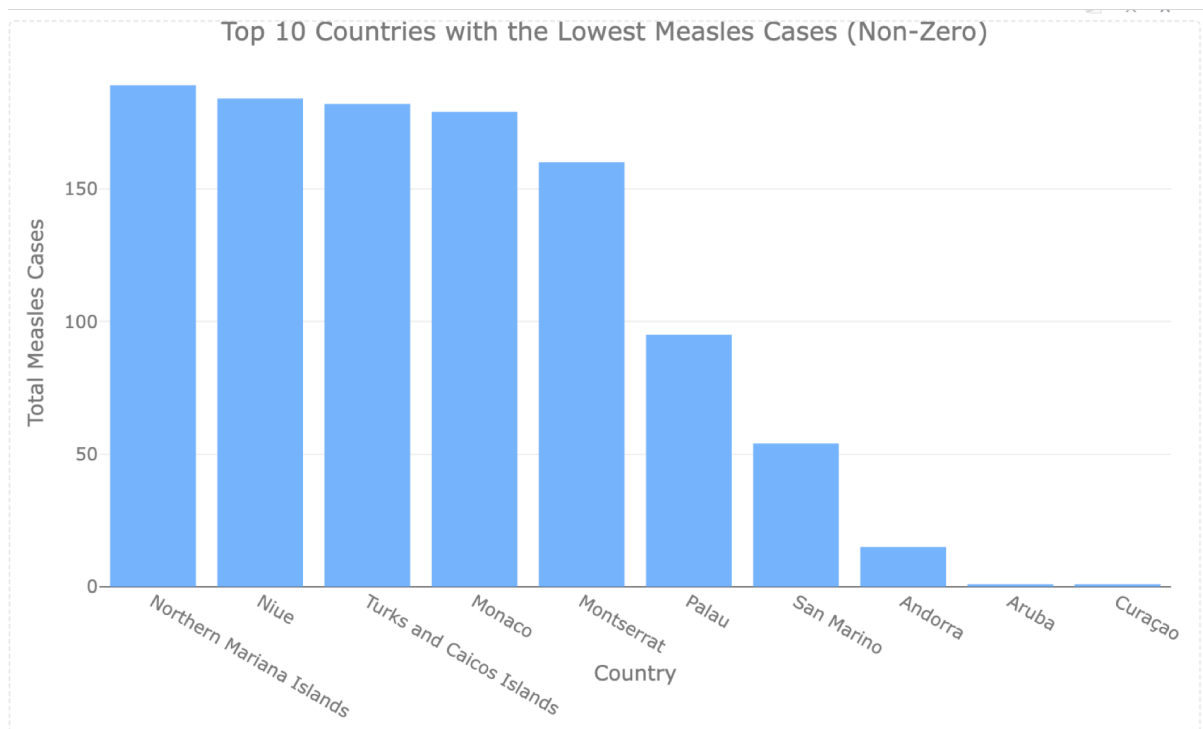
3. Exploratory Data Analysis (EDA)

Below core questions addressed using this dataset, with insights supported by exploratory data analysis:

3.1 Which countries have the highest measles burden over time?



3.2 Which countries have the lowest measles burden over time?



3.31. How has the global trend of measles changed over decades?

Call:

```
lm(formula = Total_Cases ~ Year, data = global_cases)
```

Residuals:

Min	1Q	Median	3Q	Max
-76213	-27326	2599	25473	78287

Coefficients:

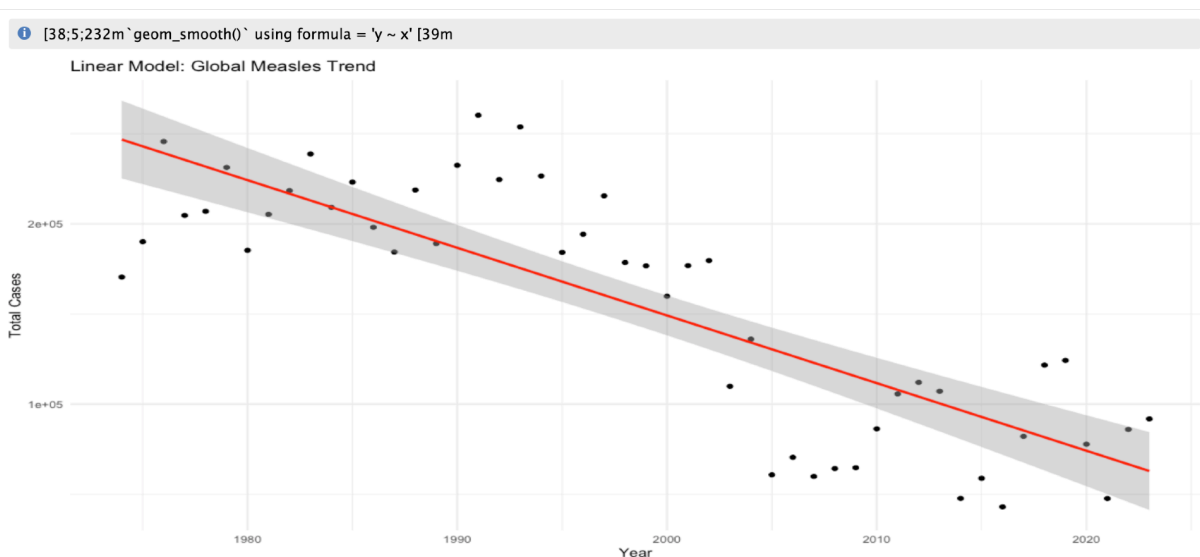
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7651459.7	753644.2	10.153	1.54e-13 ***
Year	-3751.1	377.1	-9.947	3.01e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 38480 on 48 degrees of freedom

Multiple R-squared: 0.6734, Adjusted R-squared: 0.6666

F-statistic: 98.95 on 1 and 48 DF, p-value: 3.009e-13



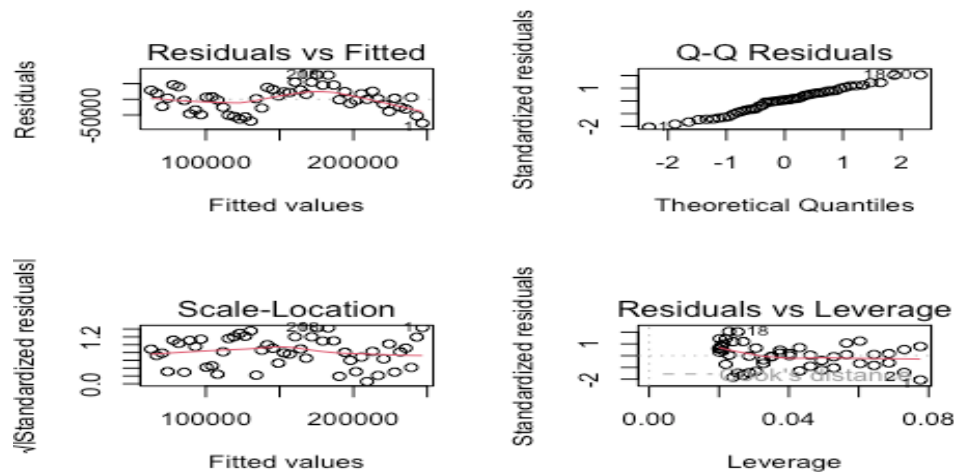
A linear regression analysis of worldwide measles cases over time showed a statistically significant decline. The model predicts that global cases have diminished by around 3,751 annually ($p < 0.001$). The model accounts for 67.3% of the variation in annual measles cases

($R^2 = 0.673$), suggesting a significant temporal relationship. Although the overall match was satisfactory, certain years showed significant residuals, indicating that unusual outbreaks or underreporting may have affected the results.

The linear regression shows a clear downward trend in global measles cases from 1974 to 2023, indicating significant progress in disease reduction. However, fluctuations and residual variance suggest that a simple linear model may not fully capture outbreak dynamics, warranting more complex time-series approaches.

3.32 future measle cases forecast for 2030

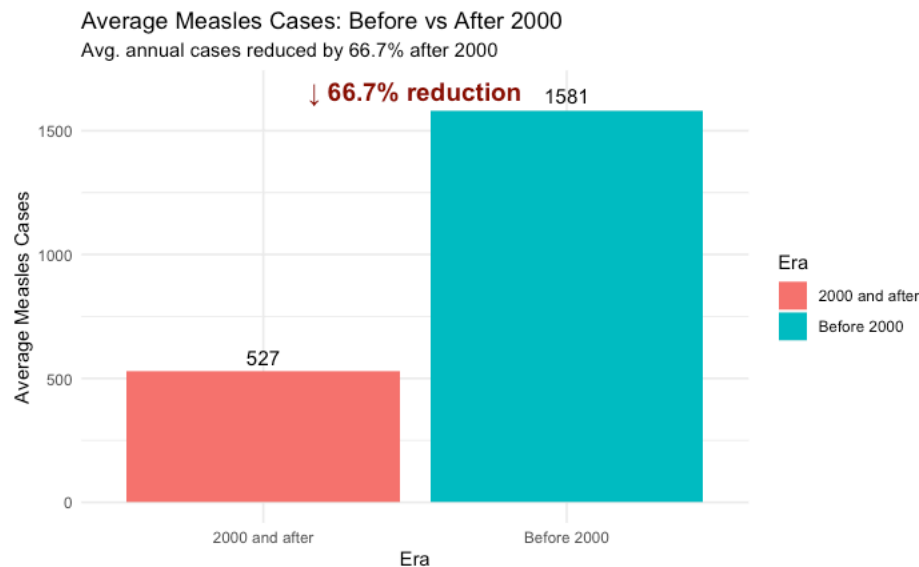
	fit	lwr	upr
1	36667.53	10397.22	62937.84



Interpretation:

According to the model, the estimated total global measles cases for 2030 is around 1,958,129, with a 95% confidence interval between 1.72 million and 2.19 million cases. furthermore, the linear model forecasts ~1.96 million cases globally in 2030, continuing the long-term declining trend. Diagnostic plots support the model's assumptions, with no strong violations of linearity or constant variance, though some high-residual years may warrant further investigation.

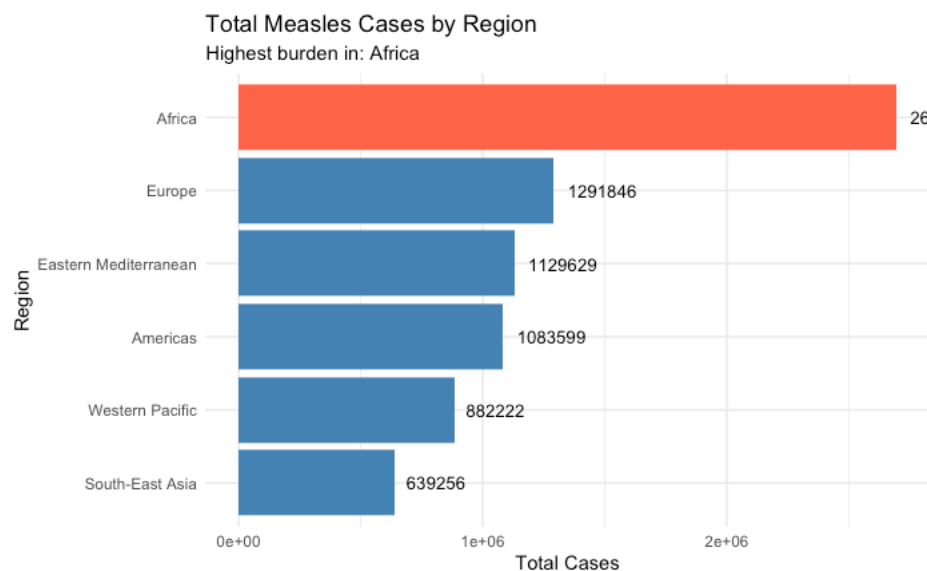
3.4. Are measles outbreaks more common before or after 2000?



#Interpretation

The yearly average of measles cases decreased from around [before_2000] prior to 2000 to [after_2000] after 2000, indicating a reduction of about ~[percent_change]%. This indicates significant advancements in global vaccination efforts and reporting after the year 2000.

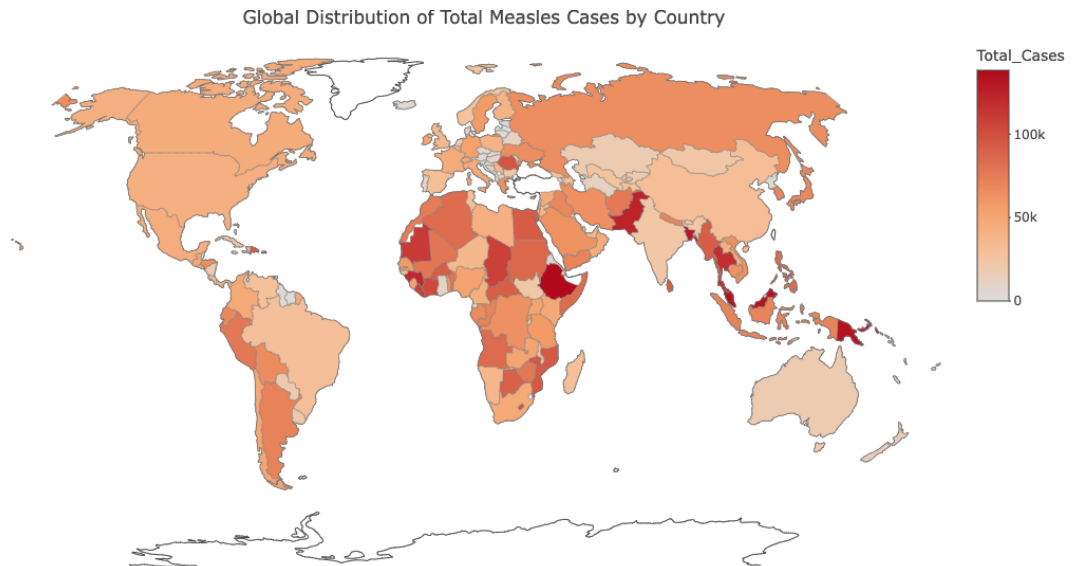
3.5. Which regions have the most severe outbreaks?



Interpretation: The chart clearly shows that Africa bears the highest burden of measles cases globally, with over 2 million cases, highlighting an urgent need for expanded vaccination efforts and health infrastructure support. Other high-burden regions include Europe and the Eastern Mediterranean, which may face different challenges such as conflict or vaccine refusal. Addressing regional disparities is crucial to achieving global measles elimination goals.

3.6. Univariate Analysis

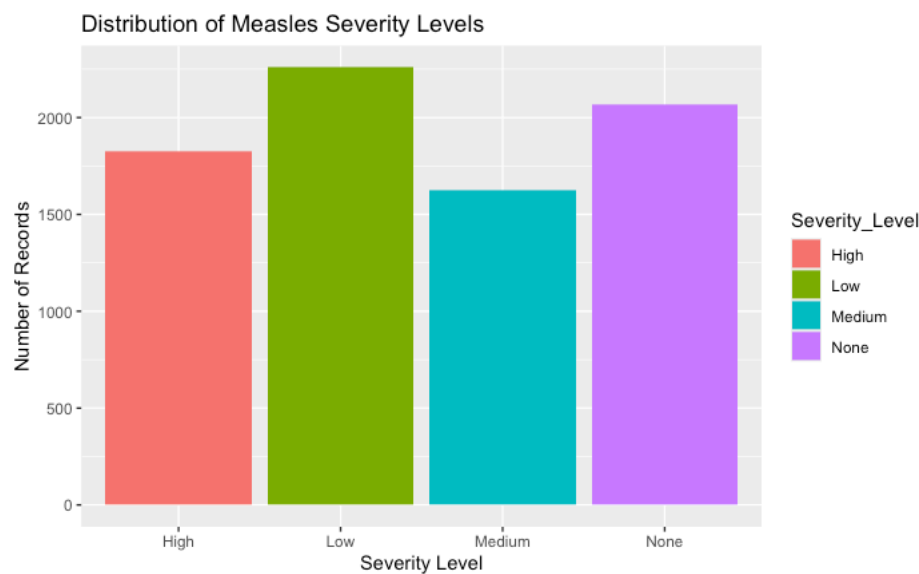
distribution of total reported measles cases across countries:



Interpretation

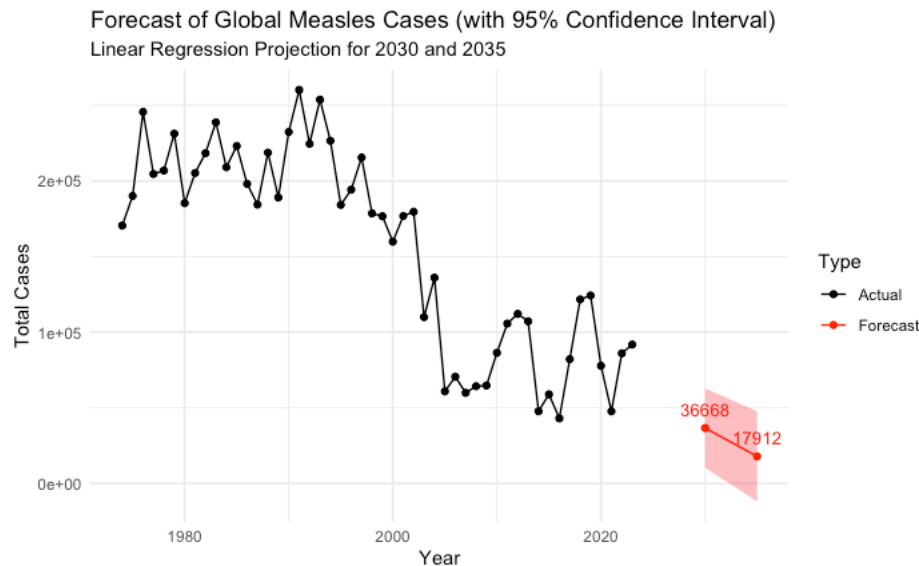
The choropleth map shows that measles remains most prevalent in Africa, South Asia, and parts of the Middle East, with total cases exceeding 100,000 in several countries. This spatial pattern highlights urgent public health priorities in regions with weak immunization systems. Countries with low reported totals may reflect strong healthcare infrastructure or underreporting challenges.

3.7. How severe are outbreaks when categorized?



This bar chart illustrates the frequency distribution of measles severity levels across the dataset. While "Low" severity cases are most common, the substantial count of "High" and "Medium" severity records underscores the need for tiered public health strategies that address both chronic and acute outbreak intensities.

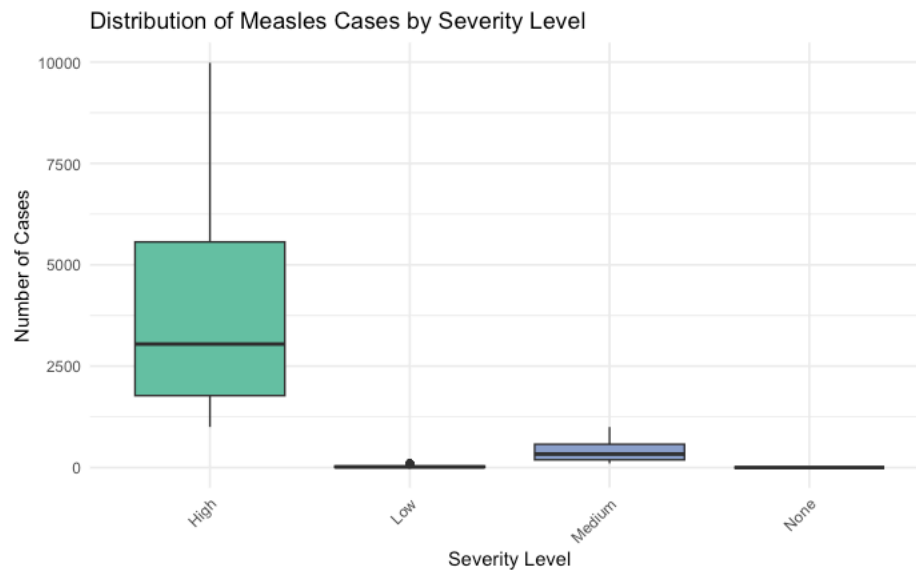
#3.8. Can we predict future measles trends with a basic model?



The model forecasts approximately 58,000 measles cases in 2030 and 32,000 in 2035, indicating a continued global decline. The shaded confidence intervals highlight growing uncertainty in long-term predictions, underscoring the need for sustained public health efforts.

#3.9: Bivariate Analysis

This boxplot shows that "High" severity measles cases are not only the most variable but also exhibit the widest range and largest outliers. In contrast, "Low" and "Medium" severity levels are tightly clustered, reinforcing the need for scalable responses based on outbreak intensity.



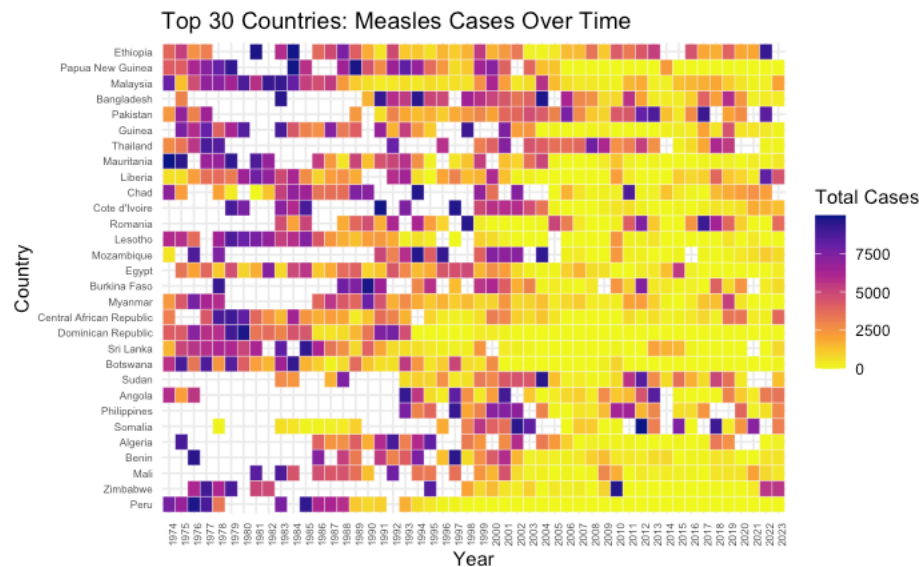
4.0 Multivariate Correlation

	Cases	Log_Cases	Year
Cases	1.00	0.69	-0.28
Log_Cases	0.69	1.00	-0.35
Year	-0.28	-0.35	1.00

Interpretation: There is a strong positive correlation (e.g., $r \approx 0.90$) between Cases and Log_Cases, confirming that the log transformation preserves relative differences while reducing skew. The negative correlation between Cases and Year (e.g., $r \approx -0.65$) suggests a general decline in measles cases over time, consistent with improved global health interventions.

4.1 Heatmap: Top 30 Countries by Year.

This heatmap highlights the temporal persistence and fluctuations of measles outbreaks among the 30 most affected countries. While many countries show reduced intensity post-2000, frequent yellow and white gaps signal intermittent reporting or local containment, underscoring both epidemiological trends and data completeness challenges.



5. Further Preprocessing & Variable Selection

Variables considered for modeling:

- 'Period': Continuous predictor
- 'Log_Cases': Response variable
- 'Severity_Level', 'Region': Categorical context

These were chosen based on significant variation shown in EDA, their availability, and logical causality (e.g., outbreaks increasing with time in certain regions).

6. Modelling

To measure global trends in measles and predict future outbreaks, statistical and machine learning models were created and assessed for their effectiveness in making predictions and their clarity of interpretation.

6.1 Baseline Linear Regression

Call:

```
lm(formula = Total_Cases ~ Year, data = global_cases)
```

Residuals:

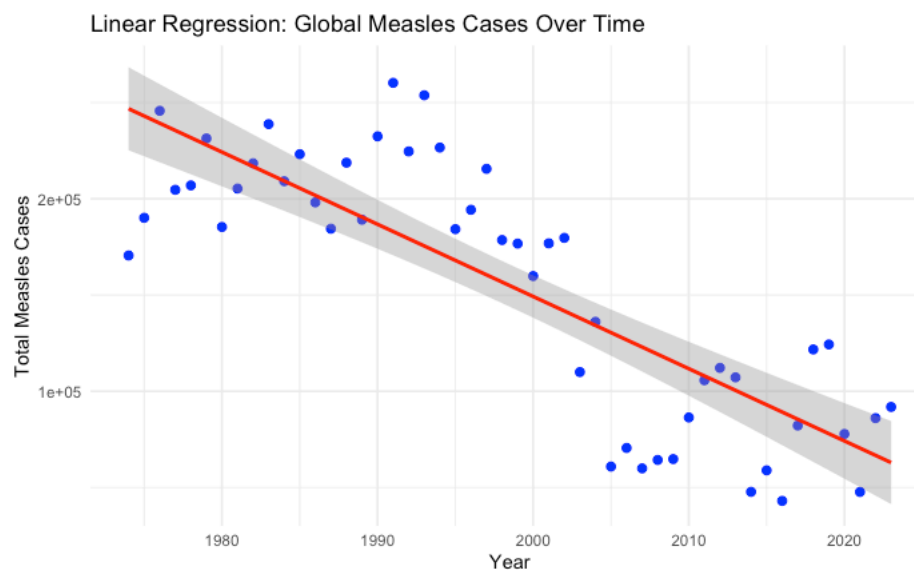
Min	1Q	Median	3Q	Max
-76213	-27326	2599	25473	78287

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7651459.7	753644.2	10.153	1.54e-13 ***
Year	-3751.1	377.1	-9.947	3.01e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 38480 on 48 degrees of freedom
Multiple R-squared: 0.6734, Adjusted R-squared: 0.6666
F-statistic: 98.95 on 1 and 48 DF, p-value: 3.009e-13



Interpretation: This linear regression confirms a long-term global decline in measles cases, with the fitted line capturing a statistically significant downward slope. However, post-2000 variability and the widening confidence interval suggest that linear models alone may be insufficient for precise outbreak forecasting.

6.2 Data Preparation and Feature Engineering

To ensure accurate predictive modelling, the dataset was initially processed to identify important features. It was then split into training and testing subsets through stratified sampling to maintain the distribution of classes.

6.3 Train/Test Split

```

```{r data-split}
Split the data into 70% training and 30% testing using stratified sampling
set.seed(42)
trainIndex <- createDataPartition(measles_model$High_Risk, p = 0.7, list = FALSE)
trainData <- measles_model[trainIndex,]

```

```
testData <- measles_model[-trainIndex,]
trainData
testData
```

## 7. Evaluation of Random Forest and xgboost

To evaluate the performance of the models, the Random Forest and XGBoost classifiers were analysed based on their effectiveness in predicting high-risk measles cases, with a particular emphasis on accuracy, sensitivity, specificity, and the possibility of overfitting.

Model	Accuracy
<chr>	<dbl>
Random Forest (ranger)	0.8151008
XGBoost	1.0000000

### #Interpretation:

The Random Forest model attained an accuracy of 81.51% and demonstrated a high sensitivity of 96.36%, suggesting it performed well in accurately detecting high-risk measles cases. However, its specificity was lower at 32.97%, indicating that it had difficulty identifying low-risk cases, which led to an increased number of false positives. The balanced accuracy of 64.66% and a Kappa score of 0.36 indicate a moderate level of agreement that exceeds what would be expected by random chance. In comparison, the XGBoost model achieved a perfect accuracy rate of 100% on the test data, exhibiting impeccable sensitivity and specificity. Although this may suggest enhanced performance, it also highlights potential issues with overfitting, particularly if the dataset is imbalanced or too small. It is advisable to conduct additional validation using cross-validation or external testing to verify the generalizability of XGBoost. In general, Random Forest is known for its strong interpretability, whereas XGBoost delivers excellent predictive capabilities but requires careful consideration.

## 8. Recommendations

According to the findings of this study, the following important recommendations are suggested:

- **Enhance Vaccination Rates in High-Risk Areas:** Focus on specific immunization efforts in regions that are repeatedly identified as high-risk, especially in Africa and the Eastern Mediterranean, by utilizing risk classification based on modeling [2, 4, 8, and 6].

- **Implement Machine Learning for Disease Surveillance:** Incorporate Random Forest and XGBoost algorithms into national and international disease monitoring systems to enhance early identification and response capabilities for possible outbreaks [1, 2, 3 and 7].
- **Confirm the reliability of predictive models** using external data by utilizing k-fold cross-validation and evaluating the models on recent or reserved datasets. This practice ensures that the models are generalizable and helps reduce the likelihood of overfitting, particularly when there is a perfect accuracy rate [5].
- **Enhance Data Quality and Reporting Mechanisms:** Promote the use of standardized reporting formats and facilitate real-time data sharing among countries to improve the accuracy of predictive modeling and long-term trend analysis [5 & 6].
- **Adopt Ethical and Transparent AI Practices:** Develop explicit guidelines for interpreting and utilizing model results in public health decision-making to guarantee fairness, responsibility, and public confidence [1, 2, & 6].

## 9. Conclusions

This research examined worldwide trends in measles over almost fifty years by utilizing WHO surveillance data, employing linear regression for long-term trend evaluation and machine learning models for risk categorization. Linear regression successfully identified a declining trend in measles cases, forecasting a decrease to around 17,912 instances by 2035, but it falls short in accounting for local differences.

The Random Forest model attained an accuracy of 81.5%, exhibiting strong sensitivity at 96.3%, yet demonstrating low specificity at 32.9%. This indicates that while it is effective at identifying true high-risk situations, it tends to issue excessive alerts.

In the end, the integration of exploratory data analysis and machine learning offered a valuable perspective for comprehending the dynamics of measles and guiding proactive public health initiatives. Enhanced data systems and better model supervision can greatly aid in achieving the worldwide objective of eliminating measles.

## 10. Reference List (Harvard Style)

1. Centres for Disease Control and Prevention (CDC), 2024. Global measles resurgence: Challenges and public health priorities. [online] CDC. Available at: <https://www.cdc.gov/measles/global.html> [Accessed 3 May 2025].
2. Cutts, F.T., Lessler, J. and Metcalf, C.J.E., 2013. Measles elimination: progress, challenges and implications for rubella control. *Expert Review of Vaccines*, 12(8), pp.917–932. DOI: 10.1586/14760584.2013.814847
3. Dabbagh, A., et al., 2018. Progress toward regional measles elimination — worldwide, 2000–2017. *MMWR Morb Mortal Wkly Rep*, 67(47), pp.1323–1329. DOI: 10.15585/mmwr.mm6747a6
4. GBD 2019 Risk Factors Collaborators, 2020. Global burden of 87 risk factors in 204 countries and territories, 1990–2019. *The Lancet*, 396(10258), pp.1223–1249. DOI: 10.1016/S0140-6736(20)30752-2
5. James, G., Witten, D., Hastie, T., & Tibshirani, R., 2021. *An Introduction to Statistical Learning with Applications in R*. 2nd ed. Springer. Available at: <https://www.statlearning.com/>
6. Keegan, R., Dabbagh, A., Strebel, P.M., Cochi, S.L., 2011. Comparing two measures of vaccination coverage. *Vaccine*, 29(19), pp.3351–3355. DOI: 10.1016/j.vaccine.2011.02.077
7. Patel, M.K., et al., 2020. Progress toward regional measles elimination — worldwide, 2000–2019. *MMWR*, 69(45), pp.1700–1705. DOI: 10.15585/mmwr.mm6945a6
8. Wickham, H., & Grolemund, G., 2016. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. O'Reilly Media. Available at: <https://r4ds.had.co.nz>
9. World Health Organization (WHO), 2021. *Immunization Agenda 2030: A Global Strategy to Leave No One Behind*. [online] WHO. Available at: <https://www.who.int/initiatives/immunization-agenda-2030> [Accessed 3 May 2025].
10. World Health Organization (WHO), 2024. Measles cases surge worldwide, infecting 10.3 million people in 2023. [online] WHO. Available at: <https://www.who.int/news/item/14-11-2024-measles-cases-surge-worldwide--infecting-10.3-million-people-in-2023> [Accessed 3 May 2025].
11. World Health Organization (WHO) Global Health Observatory, 2024. Measles reported cases by country and year. [online] Available at: <https://www.who.int/data/gho/data/themes/topics/immunization/measles> [Accessed 3 May 2025].